

K.SUJITH 2303A52346

BATCH 45

ASS 6.3

Task Description #1: Classes (Student Class)

Scenario

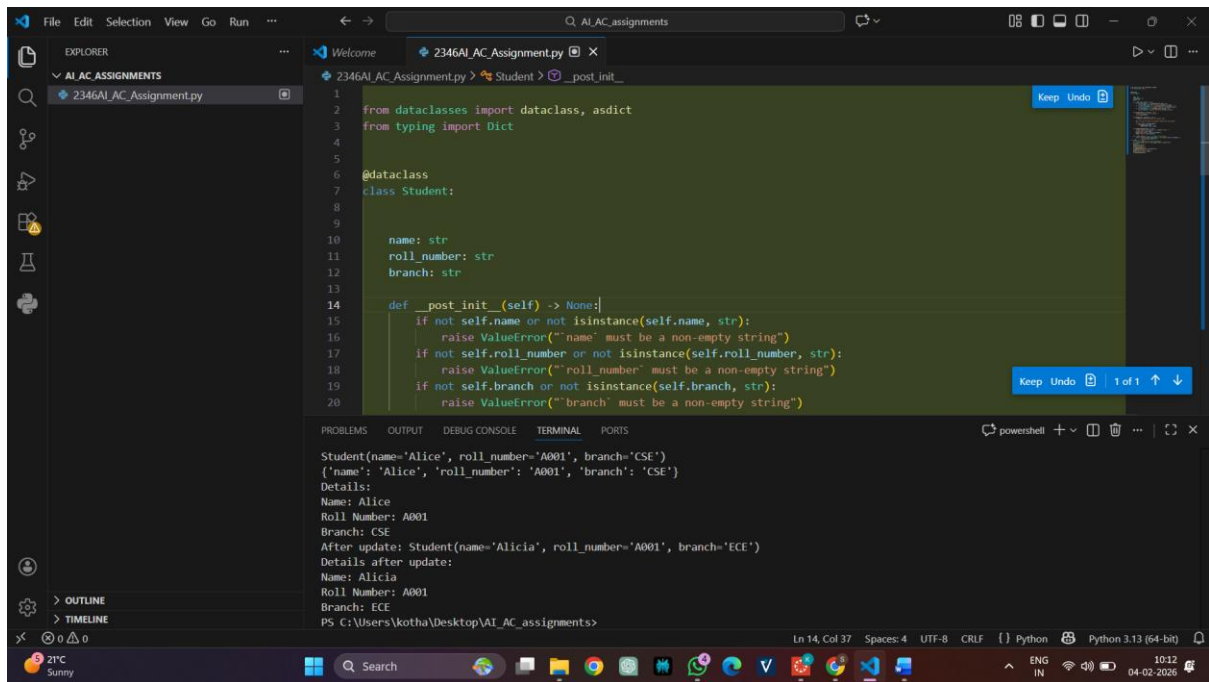
You are developing a simple student information management module.

Task

- Use an AI tool (GitHub Copilot / Cursor AI / Gemini) to complete a Student class.
- The class should include attributes such as name, roll number, and branch.
- Add a method `display_details()` to print student information.
- Execute the code and verify the output.
- Analyze the code generated by the AI tool for correctness and clarity.

Expected Output #1

- A Python class with a constructor (`__init__`) and a `display_details()` method.
- Sample object creation and output displayed on the console.
- Brief analysis of AI-generated code.



Task Description #2: Loops (Multiples of a Number)

Scenario

You are writing a utility function to display multiples of a given number.

Task

- Prompt the AI tool to generate a function that prints the first 10 multiples of a given number using a loop.

- Analyze the generated loop logic.

- Ask the AI to generate the same functionality using another controlled looping structure (e.g., while instead of for).

Expected Output #2

- Correct loop-based Python implementation.
- Output showing the first 10 multiples of a number.
- Comparison and analysis of different looping approaches

The screenshot shows a Visual Studio Code editor window with a file named '2346AI_AC_Assignment.py'. The code defines a 'Student' class with a 'branch' attribute and a 'print_multiples_for' function. The function uses a 'for' loop to print the first 10 multiples of a given number 'n'. The terminal output shows the result of running the function with 'n=3', displaying the first 10 multiples of 3 (3, 6, 9, 12, 15, 18, 21, 24, 27, 30).

```
class Student:
    def __init__(self, name, roll_number, branch):
        self.name = name
        self.roll_number = roll_number
        self.branch = branch

    def __repr__(self) -> str: # Friendly representation
        return f"Student(name={self.name}, roll_number={self.roll_number}, "
            f"branch={self.branch})"

def print_multiples_for(n: int) -> None:
    """Print the first 10 multiples of n using a for loop.

    The for-loop uses range(1, 11) so the body executes exactly 10 times and
    prints n * i for i from 1 through 10.
    """
    for i in range(1, 11):
        print(n * i)

def print_multiples_while(n: int) -> None:
    """Print the first 10 multiples of n using a while loop.
```

```
First 10 multiples of 3 (while loop):
3
6
9
12
15
18
21
24
27
30
```

Task Description #3: Conditional Statements (Age Classification)

Scenario

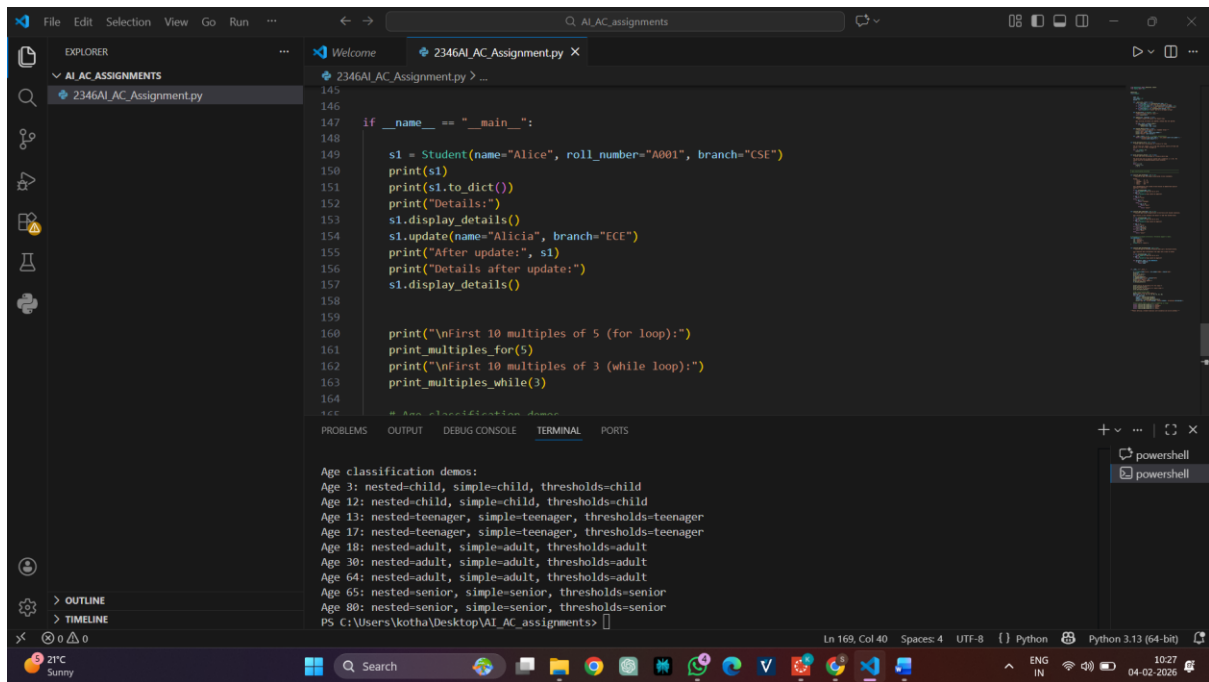
You are building a basic classification system based on age.

Task

- Ask the AI tool to generate nested if-elif-else conditional statements to classify age groups (e.g., child, teenager, adult, senior).
- Analyze the generated conditions and logic.
- Ask the AI to generate the same classification using alternative conditional structures (e.g., simplified conditions or dictionary-based logic).

Expected Output #3

- A Python function that classifies age into appropriate groups.
- Clear and correct conditional logic.
- Explanation of how the conditions work.



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145
146
147 if __name__ == "__main__":
148
149     s1 = Student(name="Alice", roll_number="A001", branch="CSE")
150     print(s1)
151     print(s1.to_dict())
152     print("Details:")
153     s1.display_details()
154     s1.update(name="Alicia", branch="ECE")
155     print("After update:", s1)
156     print("Details after update:")
157     s1.display_details()
158
159
160     print("\nFirst 10 multiples of 5 (for loop):")
161     print_multiples_for(5)
162     print("\nFirst 10 multiples of 3 (while loop):")
163     print_multiples_while(3)
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166 # Run in Terminal/Debug Console Window
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```
Age classification demos:
Age 3: nested=child, simple=child, thresholds=child
Age 12: nested=child, simple=child, thresholds=child
Age 13: nested=teenager, simple=teenager, thresholds=teenager
Age 17: nested=teenager, simple=teenager, thresholds=teenager
Age 18: nested=adult, simple=adult, thresholds=adult
Age 30: nested=adult, simple=adult, thresholds=adult
Age 64: nested=adult, simple=adult, thresholds=adult
Age 65: nested=senior, simple=senior, thresholds=senior
Age 80: nested=senior, simple=senior, thresholds=senior
PS C:\Users\kotha\Desktop\AI_AC_assignments>
```

Task Description #4: For and While Loops (Sum of First n Numbers)

Scenario

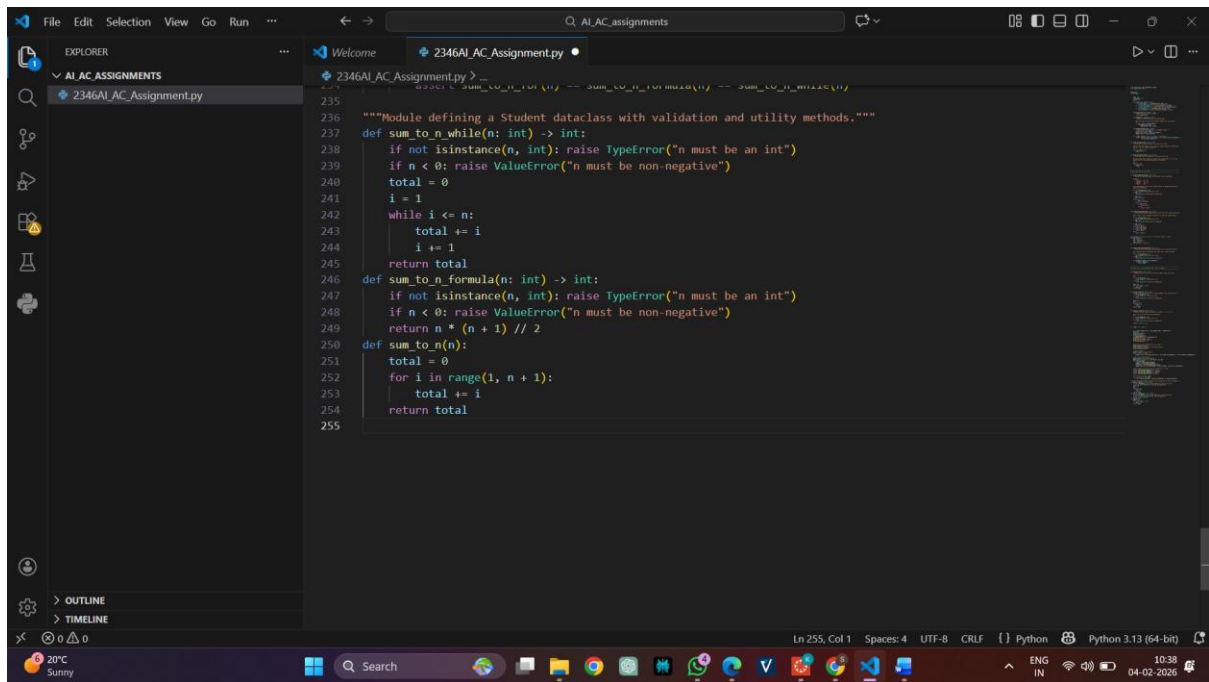
You need to calculate the sum of the first n natural numbers.

Task

- Use AI assistance to generate a `sum_to_n()` function using a for loop.
- Analyze the generated code.
- Ask the AI to suggest an alternative implementation using a while loop or a mathematical formula.

Expected Output #4

- Python function to compute the sum of first n numbers.
- Correct output for sample inputs.
- Explanation and comparison of different approaches



Task Description #5: Classes (Bank Account Class)

Scenario

You are designing a basic banking application.

Task

- Use AI tools to generate a Bank Account class with methods such as deposit(), withdraw(), and check_balance().
- Analyze the AI-generated class structure and logic.
- Add meaningful comments and explain the working of the code.

Expected Output #5

- Complete Python Bank Account class.
- Demonstration of deposit and withdrawal operations with updated balance.
- Well-commented code with a clear explanation

